Camille Vedani

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Malin Vestin, International Coordinator
School of Electrical Engineering and Computer Science (EECS)
KTH Royal Institute of Technology
SE-100 44 Stockholm, Sweden

Re: Application for a six-month study period in dependable embedded systems

Dear Malin Vestin,

I am a second-year engineering student at ENSEEIHT (Toulouse INP), in the Computer Science & Networks apprenticeship track (FISA), and an apprentice software engineer at Kyber, a French open-source start-up building infrastructure for real-time interactions. At Kyber, I develop an SDK that enables remote control of machines with less than 20 ms of latency: I implement network protocols based on QUIC and WebTransport and optimise our Rust/WebAssembly codebase for browser integration. My path is deliberately broad: two years of preparatory classes in life sciences (CPGE BCPST), a double bachelor's degree in mathematics and computer science at Aix-Marseille University, and a first master's year in embedded and connected systems at Toulouse III. Two research internships, on state reduction for probabilistic automata at LIS (CNRS) and on execution decision diagrams for worst-case execution time analysis at IRIT, taught me the rigour of academic research. Projects such as a DPLL SAT solver and model checking with Promela gave me a strong taste for formal reasoning. All of this serves one goal: designing safe, open medical embedded devices such as insulin pumps.

KTH Royal Institute of Technology stands out for its research and teaching in embedded systems with a clear focus on dependability and formal methods: exactly the combination I believe medical devices require. Its close ties with industrial leaders such as Ericsson and Scania, and its strength in medical technology, bring together the three fields I want to connect: networks, critical embedded software and health. The Swedish approach to engineering, based on collaboration, autonomy and openness, also strongly matches the way I already work at Kyber.

I would like to spend one semester (six months) at your university, within the exchange agreement currently being established between ENSEEIHT (Toulouse INP) and KTH. I would bring my skills in Rust, C/C++, Python and model checking, my experience of real-time networking, and the autonomy I have developed while working fully remotely in an international, English-speaking team. In return, this semester would let me deepen my knowledge of dependable systems and formal verification in a world-class academic environment while fulfilling the international mobility required by my engineering degree. I would be glad to provide any further information about my academic record. My CV and portfolio are attached.

Thank you for considering my application. I look forward to hearing from you.

Sincerely,

Camille Vedani